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Complex investigation of XF_3(X = Gd, Tb, Dy, Ho and Er) fluorides under pressure: An *ab-initio* perspective

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ABSTRACT

Comprehensive systematic density functional theory calculations were performed for five typical rare earth trifluorides, namely GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3, under pressures up to 30 GPa, demonstrating induced phase transitions in agreement with available experimental observations. For the first time the careful check of simulation routine is performed for the selected set of rare earth trifluorides. An extensive selection of the methodology parameters revealed different behaviors for the systems under study. Based on comparative analysis with available experimental data, suitable computation details were suggested for further calculations. For the selected trifluorides, the evolution of lattice parameters and volume, criteria of stability, and elastic stiffness coefficients were analyzed with pressure, which also were calculated for the first time. Additionally, electronic, magnetic, and optical features were captured within the scope of the work for all five compounds in two phases, along with a comparative analysis with experimental data where available.

1. Introduction

The trifluorides of rare earth elements represent a class of materials with outstanding electronic, optical, and magnetic characteristics, making them highly valuable in a broad spectrum of fields, particularly in nanoparticle form. Various studies have explored the use of these nanoparticles in computer tomography and magnetic resonance imaging, confirming their potential for biomedical applications [1–3]. Beyond nanoparticle form, the application range of rare earth trifluorides extends to laser technologies [4–7], biological labels [8], thermometry [9], drug transport systems [10], photovoltaic cells [11], hydrogen storage systems [12], catalysts [13], and matrix materials [14]. Given the focus of many practical applications and research domains on understanding the correlation between the structure of these compounds and certain physical properties, it follows that transformations and corresponding phase stability of trifluorides should be of great interest.

The orthorhombic structure with *Pnma* space group (referred to β-YF_3 type) is well-known to be adopted by GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 under ambient conditions [15]. This structure is also observed in other trifluorides of rare earth elements with smaller ionic radii from Sm to Lu, and Y [16], in contrast to trifluorides of La–Nd with higher ionic radius forming trigonal phase at ambient conditions [17]. However, there is still ambiguity related to the temperature phase behavior of these trifluorides. According to earlier studies, this family of compounds from Sm to Ho changes from orthorhombic to trigonal phase of the LaF_3 type (with $P\bar{3}c1$ space group) between 600 and 1300 K. Meanwhile, trifluorides from Er to Lu and Y change from orthorhombic to trigonal phase of the α-YF_3 type (with $P\bar{3}m1$ space group) between 900 and 1200 K in the temperature phase diagram [18–20]. However, more recent studies have revealed a more complex picture of phase transitions (PT) [21]. Specifically, trifluorides from Sm to Gd undergo a first-order PT between 750 and 1300 K to a trigonal phase of LaF_3 type. Subsequently, as the temperature increases further, they experience a second PT to a hexagonal phase of Schlyter type (of $P6_3/mcc$ space group) between 1500 and 1600 K before melting. In contrast, TbF_3, DyF_3, and HoF_3 maintain their orthorhombic structure until reaching their melting points. The last members of this family, trifluorides from Er to Lu and Y, transform to a trigonal phase of α-YF_3 type with $P\bar{3}m1$ space group between 1200 and 1450 K.

Even more limited information is available regarding the pressure-dependent phase behavior of trifluorides with small ionic radii. Concerning their initial orthorhombic phase, researchers widely acknowledge the transformation to LaF_3 type trigonal phases under external

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pressure, a phenomenon experimentally observed in compounds such as YF_3 [22], EuF_3 [23], and ErF_3 [24]. Moreover, Sui et al. recently confirmed this pressure-driven PT within the family of trifluorides using combined density functional (DFT) study and photoluminescence measurements [25]. The DFT method has proven to be effective for studying the mechanical, electronic, and thermodynamic properties of crystals under pressure, as well as for capturing pressure-induced structural phase transitions. It is also serves as a valuable tool to evaluate the stress and strain effects in materials science [26,27].

Despite the first experiments of synthesis and experimental characterization of these crystals which were performed ubiquitously [15,20], there is still a luck of comprehensive theoretical data, especially regarding the trigonal phase. Additionally, the elastic properties, as well as the magnetic features under pressure of GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 crystals have not been investigated. Moreover, experiments aimed at defining pressure-induced PT have only been conducted for GdF_3, TbF_3, and ErF_3 among investigated in present work compounds, and these researches revealed significant errors in defining transition pressures. The possibility of the existence of the hexagonal phase, corresponding to the $P6_3/mcc$ space group and identified in the temperature diagram of GdF_3 [21], has not been investigated under pressure variation conditions. Lastly, understanding the computational details and methods necessary to treat bulk rare earth trifluorides might shed light on the more complex issue of computing the properties of nanoparticles [28].

Therefore, our research aimed to perform a thorough analysis of selected trifluoride systems using *ab initio* calculations. The goal was to provide a base for experiments and further calculations involving trifluoride nanoparticles. In particular, the paper focuses on several important aspects:

1. a careful approbation of the DFT methodology, including computations with various functionals, corrections, inaccuracies, and comparisons with available theoretical and experimental data;
2. investigation the mechanical stability of orthorhombic and trigonal phases;
3. exploring the possibility of forming a hexagonal phase with the $P6_3/mcc$ space group during phase-induced transition;
4. defining the transition pressures between orthorhombic and trigonal phases, and investigating the structural and elastic properties, while also examining how these properties evolve under different pressure conditions;
5. analyzing the electronic, magnetic and optical features of both phases.

The paper is organized as follows. In Section 2, the methodology and details of the performed calculations is presented along with crystal structures. Section 3 presents the results of calculations. Initially, the structural characteristics predictions are included in Section 3.1, while Section 3.2 analyzes elastic and mechanical properties of selected compounds in both phases. Section 3.3 gives the pressure-induced PT. Section 3.4 to 3.5 analyze electronic, magnetic and optical properties. Lastly, Section 4 summarizes the performed analysis.

2. Computational details

Structural, elastic, electronic, optical and magnetic properties calculations were performed based on density functional theory [29]. To account for exchange and correlation effects, we performed calculations using the generalized gradient approximation (GGA-PBE and GGA-PBEsol) [30] in spin-polarized mode. The Kohn–Sham equations [31] were solved using projector augmented wave method (PAW) [32], as implemented in VASP-6.4 (Vienna Ab-initio Simulation Package) code [33]. The cutoff energy of the plane waves was taken to be 600 eV. The optimization of the atomic positions was carried out by the conjugate gradient method with the convergence criterion of 0.001 eV/Å, and the convergence condition for self-consistent calculations was the invariant of the total energy of the system with an accuracy of 10^{-8}, eV. Brillouin zones were sampled using Monkhorst–Pack grids [34–36], including $5 \times 5{\times}7$, $5{\times}5 \times 5$, and $3 \times 3 \times 3$ k-points centered at Gamma point, for orthorhombic, trigonal, and hexagonal cells, respectively. Gaussian smearing width was set to 0.05 eV.

We tested several approaches for describing the electronic structure of trifluorides to achieve maximum agreement with available experimental data on geometric characteristics, as well as available data containing electronic and magnetic features. First, we investigated the influence of including or excluding $4f$-electrons within the computational scheme from rare earth ions (Gd, Tb, Dy, Ho, Er) and used two types of standard PAW potentials for structure optimization. One potential considers $4f$-electrons as core states and $5p^6 5d^1 6s^2$ electrons as valence states. The other PAW potential regards $4f^x 5s^2 5p^6 5d^1 6s^2$ (where $x = 7$, 8, 9, 10, 11 for the Gd, Tb, Dy, Ho, Er ions, respectively) electrons as valence states.

Additionally, the simplified GGA+U [37] approach was applied for taking into account strong correlations between electrons. In the case of PAW potential while considering f-electrons as valence states, we used U_f values of 11.5, 5, 5.7, 6.8, and 6.9 eV for Gd, Tb, Dy, Ho and Er ions, respectively, according to Ref. [38]. When considering the PAW potential and treating $4f$-electrons as core states, we applied the Hubbard-type correction to $5d$-electrons. As a result of these tests, we found for orthorhombic GdF_3 and TbF_3 that the minimal deviation of the lattice constants and cell volumes at 0 GPa is provided by using the GGA-PBE functional and PAW potential with $4f$-electrons described as core states without any Hubbard correction. However, this scheme completely fails for DyF_3, HoF_3, and ErF_3 compounds. For these structures, we found that applying a Hubbard correction to the $5d$-electrons is crucial to accurately reproduce the experimental unit cell parameters. To determine the most suitable value of the Coulomb repulsion term for the d-electrons, a set of calculations was performed using both the GGA-PBE and GGA-PBEsol functionals with various U_d values. The results of these calculations are collected in SFigs. 1–2. This comparative analysis allowed us to select the most appropriate method to calculate the properties of the DyF_3 and HoF_3 compounds, which is the GGA-PBEsol functional with a U_d correction of 2.3 eV. This scheme has also proven to be effective for ErF_3 and is likely to be appropriate for subsequent elements of the lanthanide series above Er.

For electronic calculations, we tested both GGA-PBE and GGA-PBEsol functionals with $+U$ correction applied for d or f-electrons. For the optical properties calculations we used GGA+U_f scheme and Gaussian smearing of 0.02 width, and 432 bands involved. For calculations, we used fully relaxed structures from the previous runs. The magnetic nature of the materials was calculated using the full PAW potential including f-electrons.

The complete calculated data on the structure of the orthorhombic phases of trifluorides using different approaches along with experimental data and percentage deviations from experimental cell parameters are presented in the Supplementary Materials (Tables 1–5).

The optimized unit cells for all the considered phases are demonstrated in Fig. 1 and SFig. 2.

3. Results

3.1. Structural properties

In the low-pressure phase, GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 trifluorides exhibit an orthorhombic structure with the *Pnma* space group and four formula units per unit cell. Upon transformation to the high-pressure phase, they adopt a trigonal structure. As discussed in the Introduction, two kinds of trigonal and one type of hexagonal phases are possible for rare-earth trifluorides with small ionic radii in their temperature phase diagram at ambient pressure. These are the $P\bar{3}c1$, $P6_3/mcc$, and $P\bar{3}m1$ space groups, with six, two and one formula units per unit cell, respectively. However, keeping as reference recent experiments on phase-induced transitions in GdF_3, TbF_3 [25], and

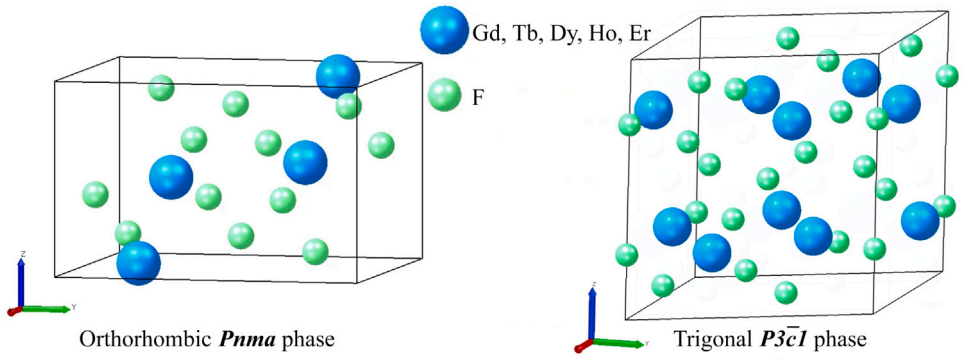


Fig. 1. Unit cells of XF_3 (X = Tb, Gd, Dy, Ho, and Er) with orthorhombic (*Pnma* space group) and trigonal ($P\bar{3}c1$ space group) structures.

ErF_3 [24], which show the formation of trigonal structures with the $P\bar{3}c1$ space group (LaF_3 type), we also modeled hexagonal structures of the Schlyter type (with the $P6_3/mcc$ space group) as possible formation since this type was identified for GdF_3 in its temperature phase diagram. However, we found that the LaF_3 type of structure is more energetically favorable for all studied trifluorides, as will be discussed in Section 3.3.

It is worth noting that low-pressure states do not necessarily imply the existence of the trigonal phases, and conversely, high-pressure states do not lead to the formation of the orthorhombic phases. In fact, to our knowledge, there is only one report on HoF_3 about synthesis in powder form, where the phase composition was identified as a mixture of trigonal and orthorhombic phases at ambient conditions [15]. However, in order to provide a comprehensive study of the properties of GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 crystals, all calculations were conducted for both phases within the pressure range of 0–30 GPa. The calculated lattice parameters and cell volumes at 0 GPa exhibit notable agreement with the available experimental data (as collected in Table 1). Moreover, experimental lattice parameters of trigonal ErF_3 at 19.6 GPa, a = b = 6.56 Å, c = 6.70 Å [24], are in excellent agreement with our PBEsol $+U_d$ calculations at 20 GPa, a = b = 6.49 Å, c = 6.69 Å. This alignment underscores the reliability of the chosen calculation method.

The evolution of the lattice parameters for the unit cells of GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 in both the orthorhombic and trigonal phases with applied pressure is illustrated in Fig. 2. Linear compressibilities, expressed as $-\frac{1}{L}(\frac{dL}{dp})_T$, can be evaluated from variations in lattice parameters data. The results will be presented in the next section in Table 3. Significant anisotropy in values, corresponding to different crystal axes for orthorhombic phases is observed there.

The variations of the calculated unit cell volumes for the studied trifluorides in both the orthorhombic and trigonal phases are depicted in Fig. 3. This data allows to determine the bulk modulus B_0 by fitting the calculated volume-pressure curve to a fourth-order Birch–Murnaghan equation of state [41]. The obtained equation parameters are collected in Table 6. In the next section, these parameters will be subsequently compared with those obtained from elasticity constants (Table 3 of the next section). Available in the literature experimental bulk modulus for ErF_3 [24] is 130 GPa for the orthorhombic phase, which is in a good agreement with our calculation result of 132.38 GPa. Unfortunately, the comparison with experimental data for other compounds cannot be performed due to absence of it.

3.2. Elastic constants and mechanical properties

Studying the elastic properties of materials is essential for understanding their response to pressure and evaluating the stability of their structure. Additionally, elastic constants offer valuable insights into the bonding characteristics between adjacent atomic planes and the anisotropic nature of the bonding, while also playing a pivotal role in determining the mechanical properties of solids.

Table 1
Optimized lattice constants *a*, *b* and *c* (all in Å) and unit cell volumes *V* (in Å^3) of the XF_3 (X = Gd, Tb, Dy, Ho and Er) at 0 GPa comparing to the experimental data.

		a	*b*	*c*	*V*
Pnma					
GdF_3	This work GGA-PBE	6.52	6.99	4.45	202.7
	Exp. Ref. [15]	6.57	6.98	4.39	201.6
	Exp. Ref. [25]	6.55	6.98	4.40	201.4
TbF_3	This work GGA-PBE	6.44	6.96	4.46	199.9
	Exp. Ref. [15]	6.51	6.95	4.38	198.4
	Exp. Ref. [20]	6.51	6.98	4.39	199.5
DyF_3	This work GGA-PBEsol+U_d	6.48	6.93	4.36	195.8
	Exp. Ref. [15]	6.46	6.91	4.38	195.2
	Exp. Ref. [20]	6.46	6.91	4.38	195.5
HoF_3	This work GGA-PBEsol+U_d	6.42	6.89	4.35	192.7
	Exp. Ref. [15]	6.40	6.88	4.38	192.8
	Exp. Ref. [39]	6.40	6.88	4.38	192.8
	Calc. Ref. [40]	6.42	6.88	4.34	191.8
ErF_3	This work GGA-PBEsol+U_d	6.38	6.86	4.35	190.3
	Exp. Ref. [15]	6.35	6.85	4.38	190.5
	Exp. Ref. [24]	6.34	6.84	4.39	190.5
$P\bar{3}c1$					
GdF_3	This work GGA-PBE	6.90	6.90	7.08	291.3
TbF_3	This work GGA-PBE	6.86	6.86	7.04	286.8
DyF_3	This work GGA-PBEsol+U_d	6.82	6.82	6.99	281.6
HoF_3	This work GGA-PBEsol+U_d	6.78	6.78	6.95	277.1
	Exp. Ref. [15]	6.83	6.83	6.98	282.4
ErF_3	This work GGA-PBEsol+U_d	6.76	6.76	6.92	273.7

As described in Section 3.1, crystals of GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 exhibit low-pressure *Pnma* orthorhombic and high-pressure $P\bar{3}c1$ trigonal structures. Therefore, these phases are characterized by nine (C_{11}, C_{12}, C_{13}, C_{22}, C_{23}, C_{33}, C_{44}, C_{55}, C_{66}) and seven (C_{11}, C_{12}, C_{13}, C_{14}, C_{15}, C_{33}, C_{44}) independent elastic constants, respectively [42].

The calculated values of elastic constants at 0 GPa for the trifluorides studied in the present research, in both orthorhombic and trigonal phases, are presented in Table 2 (the C_{15} constants for the trigonal phase were zero for all the compounds; therefore, they are not included). To our knowledge, there is only one paper reporting theoretical data on elastic constants for orthorhombic HoF_3 [43]. However, these calculations were performed using the GGA-PBE approach, which, as we found in our test calculations, does not provide accurate cell parameters being a crucial factor for defining the elastic behavior of materials. Therefore, we cannot rely on this data as a reference. Since there are no other theoretical or experimental results available for the structures under investigation in the literature, we anticipate that our predictions should be checked in future experimental researches on these materials.

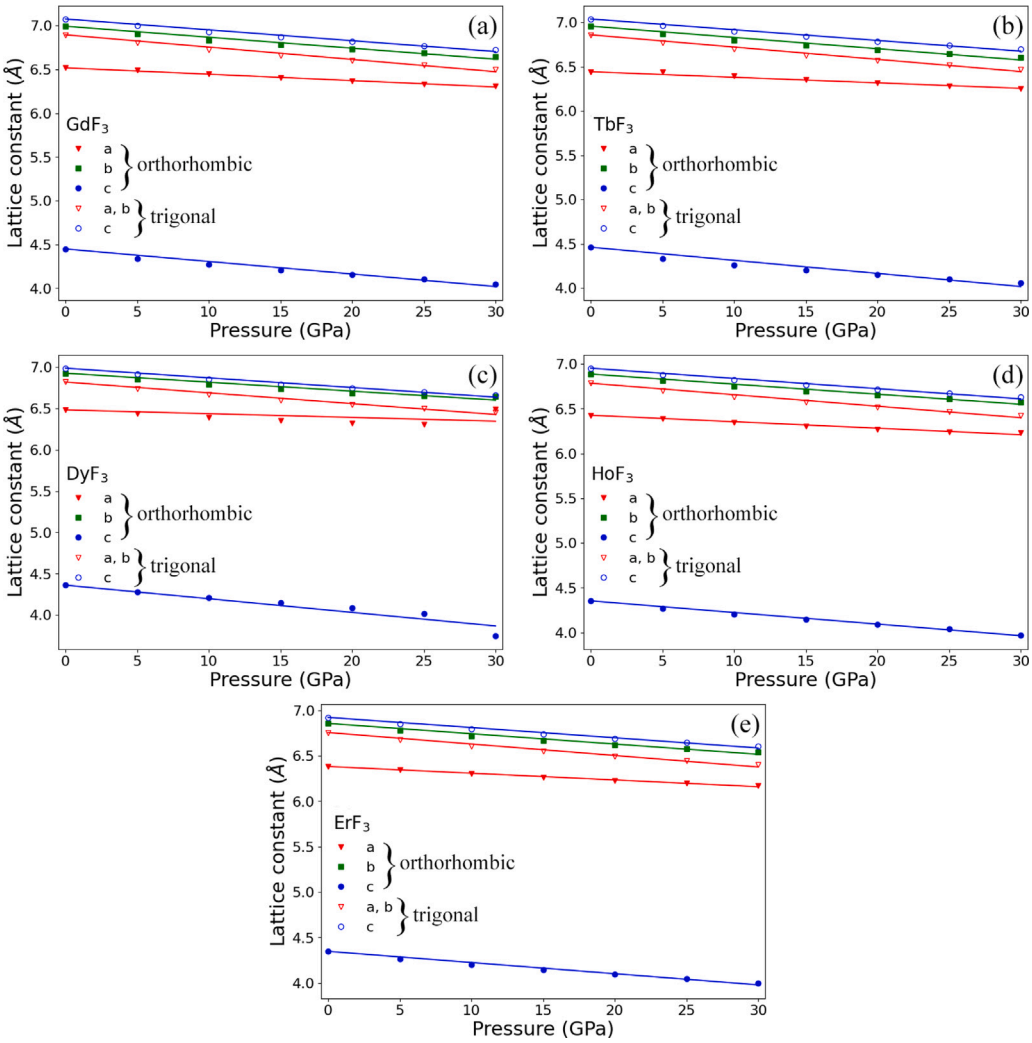


Fig. 2. Calculated evolution of lattice parameters with pressure of GdF_3 in GGA-PBE (a), TbF_3 in GGA-PBE (b), DyF_3 in GGA-PBEsol+U_d (c), HoF_3 GGA-PBEsol+U_d (d), and ErF_3 GGA-PBEsol+U_d approaches (e). Red, green and blue solid lines denote fitting results to linear function.

Table 2
Calculated elastic constants C_{ij} (all in GPa) for the orthorhombic and trigonal XF_3 (X = Gd, Tb, Dy, Ho, and Er) structures at 0 GPa.

Pnma	C_{11}	C_{12}	C_{13}	C_{22}	C_{23}	C_{33}	C_{44}	C_{55}	C_{66}
GdF_3	195.57	83.16	97.81	217.24	55.68	131.61	53.05	53.44	60.56
TbF_3	200.97	90.06	123.13	227.84	78.24	147.72	53.27	56.48	60.37
DyF_3	213.59	90.68	111.57	246.22	61.60	153.83	58.36	56.92	66.79
HoF_3	218.31	90.57	116.01	248.37	64.26	151.18	61.74	59.10	68.56
ErF_3	221.31	93.62	119.35	249.46	62.23	151.47	63.37	60.93	68.70

$P\bar{3}c1$	C_{11}	C_{12}	C_{13}	C_{14}	C_{33}	C_{44}
GdF_3	200.81	100.57	76.91	0.55	246.63	42.27
TbF_3	196.82	98.95	78.01	1.16	244.96	42.10
DyF_3	200.50	86.82	58.13	−3.10	246.99	45.99
HoF_3	207.59	100.32	73.51	−2.94	268.87	47.30
ErF_3	214.04	103.05	76.89	−2.95	277.42	51.26

The elastic constants C_{11}, C_{22}, and C_{33} are associated with unidirectional compression along the principal crystallographic directions, while C_{44}, C_{33}, and C_{66} are responsible for resistance against shear deformations. Based on the calculated values of the elastic constants, significantly larger values of C_{11}, C_{22}, and C_{33} are exhibited by GdF_3 and TbF_3, DyF_3, HoF_3, and ErF_3 in the orthorhombic phase, demonstrating greater resistance to unidirectional compression than to shear deformation. The same behavior is observed in the trigonal phase. The bulk modulus provides insight into the mechanical properties of materials under hydrostatic pressure. This parameter characterizes a material's resistance to deformation and can be calculated using the Voigt–Reuss–Hill approach based on the elastic constants. By employing standard relations tailored for orthorhombic and trigonal crystals, we have determined the bulk moduli of the studied phases at 0 GPa, given in Table 3. A good agreement between the bulk moduli calculated by different methods is observed. Moreover, there is an excellent agreement with the available experimental data of the orthorhombic and trigonal ErF_3.

As pressure applied to a crystal, the way a material responds, changes. Instead of relying on the elastic constants C_{ij}, the elastic stiffness coefficients B_{ij} are used. When the stress aligns with hydrostatic pressure p, these coefficients for a crystal are represented as $B_{ii} = C_{ii} - p$ for diagonal and $B_{ij} = C_{ij} + p$ for non-diagonal coefficients, where C_{ij} are the elastic constants calculated for a structure under current pressure [44,45]. Calculated variations of the elastic stiffness coefficients with pressure for the orthorhombic and trigonal phases of GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 crystals are presented in SFig. 4 and SFig. 5 for orthorhombic and trigonal phases, respectively. We found, that all the compounds in a given phase exhibit very similar trends in the variation of stiffness coefficients with pressure, differing only in their specific values. This indicates that all the trifluorides respond to external pressure in the same way.

As the next step, we examined the mechanical stability of the phases under pressure. When subjected to hydrostatic pressure, stability

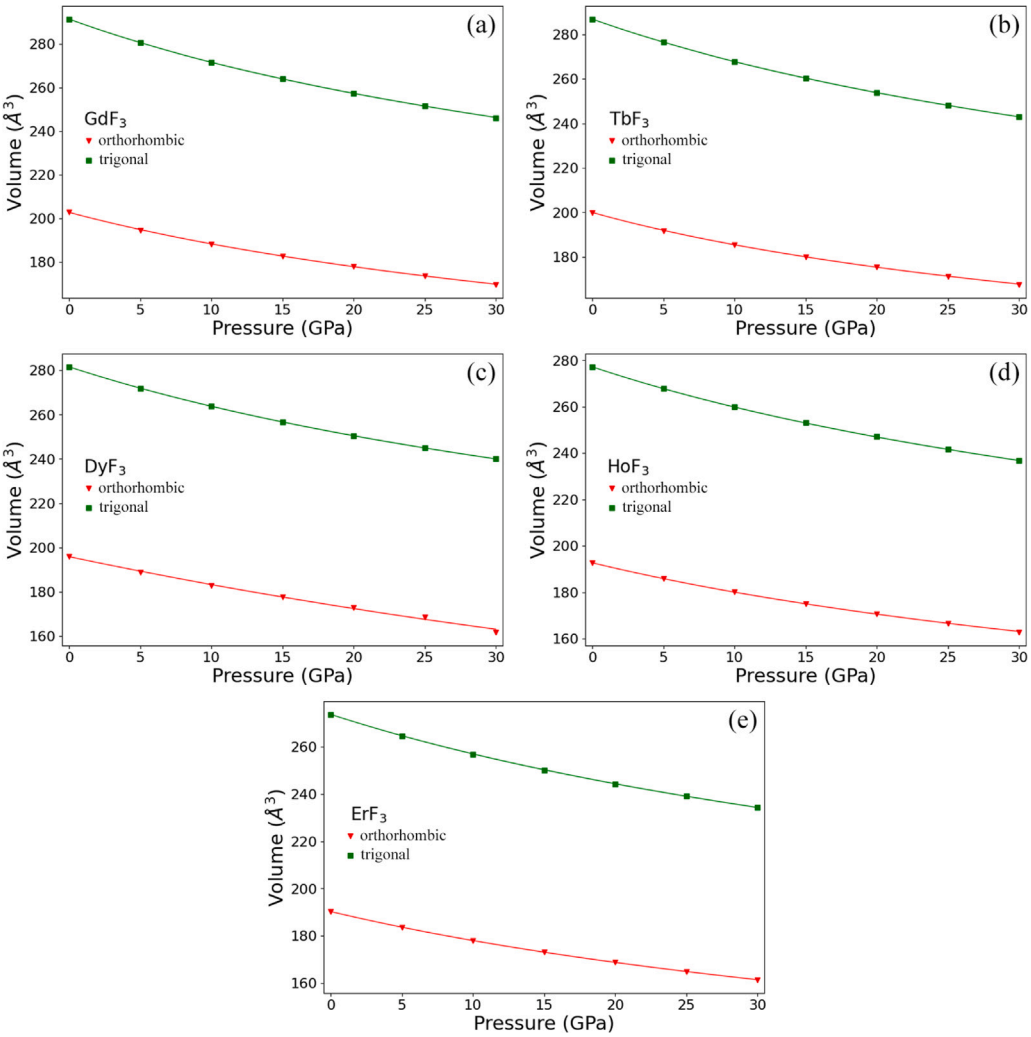


Fig. 3. Calculated volumes of GdF_3 in GGA-PBE (a), TbF_3 in GGA-PBE (b), DyF_3 in GGA-PBEsol+U_d (c), HoF_3 in GGA-PBEsol+U_d (d), and ErF_3 in GGA-PBEsol+U_d approaches (e) as a function of pressure. Red and green solid curves denote fitting results to the fourth-order Birch–Murnaghan equation of state with obtained fitting parameters collected in Table 6.

Table 3
Linear bulk compressibilities K_i along the x, y, z axes (all in TPa^{-1}) obtained from cell parameters, and bulk moduli (all in GPa): B_0 obtained by fitting the calculated volume-pressure curve to a fourth-order Birch–Murnaghan equation of state, B_V, B_R, B_{VRH}, calculated using the Voigt, Reuss, and Hill approximations from the calculated elastic coefficients for the orthorhombic and trigonal XF_3 (X = Gd, Tb, Dy, Ho, and Er) at 0 GPa. Bulk moduli for the trigonal ErF_3 at 20 GPa was estimated from elastic constants as well for comparison with available experimental data.

Pnma	K_a	K_b	K_c	B_0	B_V	B_R	B_{VRH}	
GdF_3	1.16	1.80	3.22	116.35	113.10	105.59	109.33	
TbF_3	0.97	1.83	3.32	110.87	128.82	123.92	126.37	
DyF_3	0.70	1.57	3.78	128.74	126.81	120.39	123.60	
HoF_3	1.13	1.64	2.98	129.35	128.84	121.29	125.06	
ErF_3	1.16	1.65	2.83	132.38	130.29	121.53	125.91	130 [24]
$P\bar{3}c1$	K_a	K_c	B_0	B_V	B_R	B_{VRH}		
GdF_3	2.04	1.75	122.35	128.56	128.34	128.45		
TbF_3	2.02	1.72	126.07	127.61	127.27	127.44		
DyF_3	1.92	1.67	133.95	130.97	130.49	130.73		
HoF_3	1.89	1.64	134.81	133.57	133.11	133.34		
ErF_3	1.87	1.62	139.93	135.46	134.92	135.19		
ErF_3 (20 GPa)				206.62	206.26	206.44	208 [24]	

conditions, known as the generalized Born criteria, must be assessed through the elastic stiffness coefficients [46,47]:

$$M_1 = B_{11} > 0; M_2 = B_{11}B_{22} - B_{12}^2 > 0;$$
$$M_3 = B_{11}B_{22}B_{33} + 2B_{12}B_{13}B_{23} - B_{11}B_{23}^2 - B_{22}B_{13}^2 - B_{33}B_{12}^2 > 0;$$
$$M_4 = B_{44} > 0; M_5 = B_{55} > 0; M_6 = B_{66} > 0$$

for the orthorhombic cell, and

$$M_1 = B_{11} - |B_{12}| > 0; M_2 = B_{33}(B_{11} + B_{12}) - 2B_{13}^2 > 0;$$
$$M_3 = B_{44}(B_{11} - B_{12}) - 2B_{14}^2 > 0$$

for the trigonal structure.

For all structures, the stability criteria are fulfilled in the entire 0–30 GPa range, suggesting mechanical stability under these conditions for both orthorhombic and trigonal phases (SFigs. 6, 7).

3.3. Pressure-induced phase transition

From the calculations, one can predict the transition pressure by comparing the enthalpy as a function of pressure for different phases using the equation $\Delta H = (U_t + pV_t) - (U_o + pV_o)$, where U represents internal energy obtained from the *ab-initio* calculations, V is the cell volume, p is the pressure, and o and t refer to the orthorhombic and trigonal phases, correspondingly. The transition from the orthorhombic to

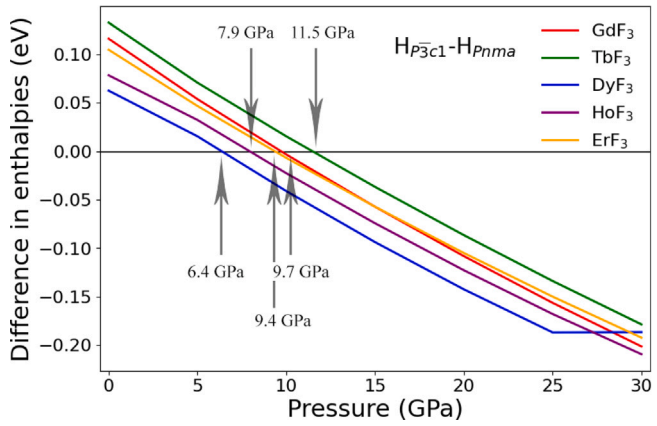


Fig. 4. The GGA-PBEsol+U_d calculated variation in enthalpy difference (ΔH) between trigonal and orthorhombic phases with $P\bar{3}c1$ and *Pnma* space groups for GdF_3 (red), TbF_3 (green), DyF_3 (blue), HoF_3 (purple), and ErF_3 (orange) crystals with pressure. Transition pressures are indicated with arrows.

trigonal phase occurs at the pressure where the ΔH function intersects the abscissa axis.

First, we compared enthalpies per cell formula unit between trigonal and hexagonal phases with $P\bar{3}c1$ and $P6_3/mcc$ space groups, respectively. The variation in enthalpy difference between these two phases with pressure is shown in SFig. 8. We found that the structure with trigonal $P\bar{3}c1$ space group is energetically more favorable. Therefore, we could not confirm the formation of the hexagonal phase with the $P6_3/mcc$ space group in the 0–30 GPa range for all the studied compounds. However, since DFT does not include temperature effects, our finding is not addressed for ambient conditions.

The obtained results on variation in enthalpy difference between trigonal and orthorhombic phases ($P\bar{3}c1$-*Pnma*) with pressure for GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3 crystals are presented in Fig. 4. From the zero cross it follows that the pressures corresponding to the phase induced transition between orthorhombic and trigonal phase with $P\bar{3}c1$ space group are 9.7, 11.5, 6.4, 7.9, and 9.4 GPa for GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3, respectively.

Let us remind that we used different approaches for the first two and the last three compounds in order to match experimental lattice parameters. Consequently, the transition pressures obtained via GGA-PBE functional for GdF_3 and TbF_3 were found to be at the upper limit of the experimental range reported in Ref. [25]. Besides, these values align perfectly with the calculated values published in that study. At the same time the GGA-PBEsol functional and +U_d correction, which we proposed as the most suitable method for DyF_3, HoF_3, and ErF_3, results in a substantial discrepancy compared to the calculated transition pressures in that study. In particular, the differences between the transition pressures calculated by GGA-PBE and GGA-PBEsol +U_d for DyF_3, HoF_3, and ErF_3 are 6.9, 7.2 and 7.8 GPa, respectively (GGA-PBE calculated variation in enthalpy difference between trigonal and orthorhombic phases with pressure is presented in SFig. 9). Moreover, for the ErF_3, the pressure calculated by GGA-PBE is 17.2 GPa [25], which is above the experimental range of 9.4–16.1 GPa [24]. The comparison of these results with the available experimental transition range give us the reason to claim that GGA-PBEsol is more precise and provides values within the limit of the experimental range.

Lastly, the corresponding volume shrinkage values during the PT were evaluated as 4.0, 3.8, 4.4, 4.3, and 3.9% for GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3, respectively.

3.4. Electronic and magnetic properties

For further investigation of trifluoride crystals, we analyzed their electronic characteristics by examining the density of states (DOS) and the evolution of band gaps under varying pressure conditions. The obtained DOSes for all investigated compounds in both phases exhibit similar features, indicating insulating behavior. The fluorine (F) states predominantly appear at the top of the valence band, while the rare-earth *d*-electrons occupy the bottom of the valence band. We plot the DOS only for DyF_3 since all others show similar features, differing only in the size of the band gap (see SFig. 10). When comparing various methods within the PAW approach used in VASP, it appears that neither GGA-PBE nor GGA-PBEsol, nor +U corrections are able to fully capture all the electronic properties of these strongly correlated materials containing rare earth ions.

Table 4
The calculated within GGA-PBE and GGA-PBEsol+U_d band gaps (ϵ) and magnetic moments (μ_B) per rare-earth ion for investigated compounds in the orthorhombic phase along with available in literature experimental data and DFT calculations.

		GGA-PBE		GGA-PBEsol		
		GdF_3	TbF_3	DyF_3	HoF_3	ErF_3
ϵ, eV	This work	7.76	7.76	8.07	7.75	7.57
	Theory	6.9 [48]		7.5 [48]	8 [40]	
	Exp.	8.0 [49]		7.9 [50]		
				8.6 [49]		
μ_B	This work	6.966	5.886	4.892	3.898	2.930
	Theory	6.986 [48]	5.99 [51]	4.984 [48]		
	Exp.		9.96 [52,53]	8 [54]	5.7 [55]	

Indeed, in the DOSes without *f*-electrons, the band gaps are correctly represented and consistent with experimental evidence. However, the magnetic nature of the materials is not captured since only valence *f*-electrons are responsible for the ferromagnetism in these compounds. Contrary, including *f*-electrons changes the magnetic distribution drastically and ferromagnetic nature restores, but the position of *f*-states in the band gap is not physically correct, leading to an inaccurate evaluation of the band gap. Moreover, the improper positioning of the *f*-states affects the entire energy distribution near the Fermi level, resulting in inaccurate description of both the valence and conduction bands. One possible solution is to use other approaches, including full potentials, as utilized in Refs. [48,51].

The comparison of the band gaps calculated using GGA-PBE and GGA-PBEsol with available DFT and experimental values is presented in Table 4 for the orthorhombic phase of the investigated compounds. The good agreement with experimental band gaps confirms that the proposed optimization and calculation procedure is well-suited for this purpose. Moreover, the used set of parameters results in better agreement with experiments than the full potential approaches used in Refs. [48,51]. However, as shown in the table, the description of magnetic moments remains far from the experimental values for both methods.

Next, in order to analyze the pressure dependencies of insulating properties, we plot the energetic positions of the top of the valence band and the bottom of the conduction band in Fig. 5, as well as the evolution of band gaps versus pressure in the inset. We can see that pressure changes the distribution of energy levels, causing an upshift of the corresponding levels for both phases. Despite these changes, the insulating behavior is preserved across the entire pressure range, as expected.

To sum up, while considering the electronic structure, we confirmed the findings previously published in Ref. [48] that the PAW potential, treating 4*f*-electrons as valence states, is not capable of accurately capturing both magnetic and band gap characteristics. In particular, the energy position of *f* states is in the middle of the band gap. Neither GGA-PBEsol, nor +U_f or +U_d provided any improvement in this regard. In order to overcome such limitations, we used GGA-PBE with *f*-electrons in core to describe electronic characteristics, as captured by density of states plots, and the functional with *f*-electrons in valence to calculate magnetic moments. However, none

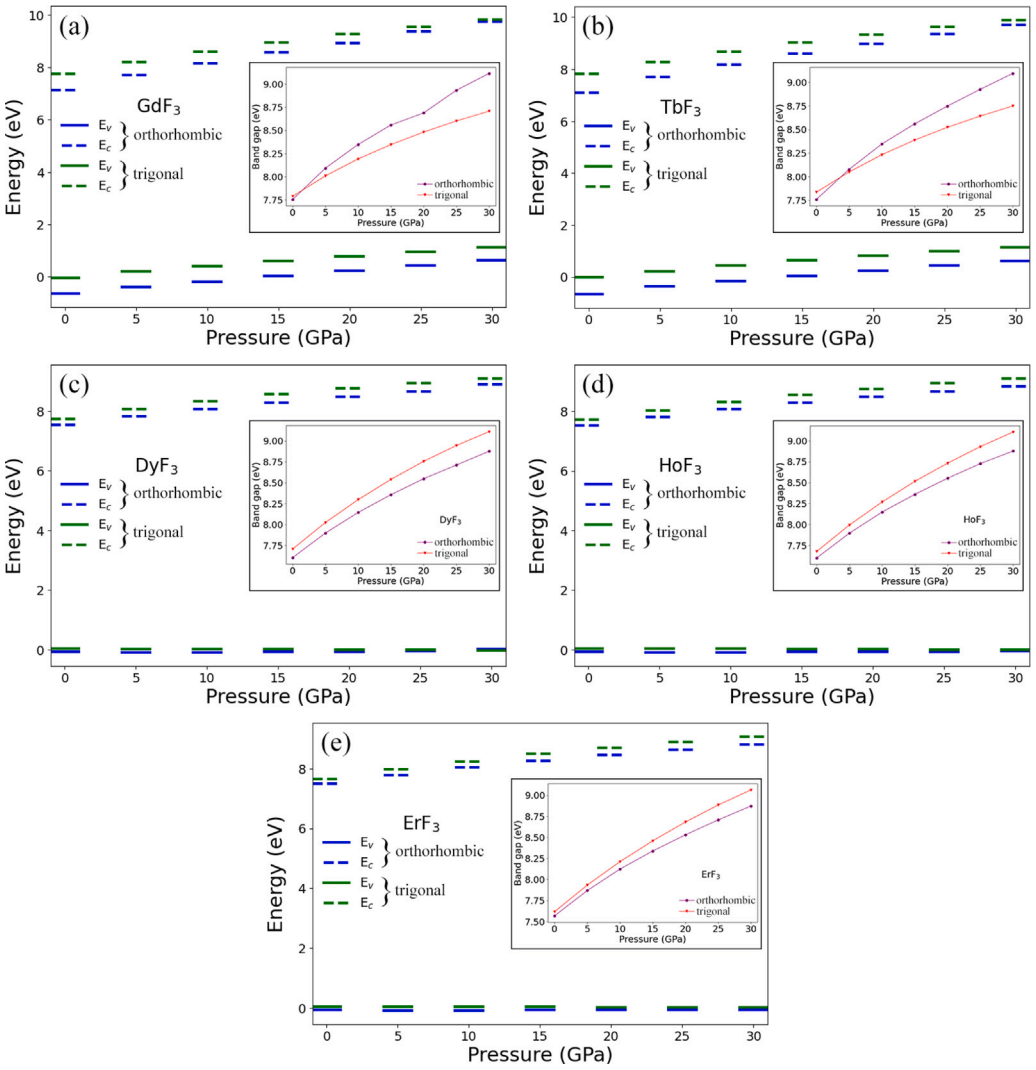


Fig. 5. Energy diagrams, showing the evolution in energies of the upper edge of the valence band and the lower edge of the conduction band, and the corresponding band gaps for GdF_3 (a), TbF_3 (b) (calculated using GGA-PBE functional), DyF_3 (c), HoF_3 (d), and ErF_3 (e) (calculated in GGA-PBEsol+U_d approach) with pressure. The Fermi level aligns with the upper edge of the valence bands.

of the tested approaches accurately described the magnetization of the compounds, and all significantly underestimated the experimental magnetic moments, although the qualitative behavior was well coincided.

3.5. Optical spectra

The last set of calculations was performed to analyze optical characteristics, specifically the reflectivity indexes. The resulting spectra for all investigated crystals in both phases at 0 GPa are collected in Fig. 6, along with available experimental data.

Recently, Saini et al. [48] performed similar calculations for a range of compounds, namely trifluorides of La, Ce, Pr, Nd, Gd and Dy using a full potential approach. Similar to that research, we found no visible difference by including the U_d parameter in optical properties calculations, except for small deviations at low energies. Nevertheless, the presented reflectivity dependencies were computed with the additional Coulomb term. The resulting spectra features as transparency in the low-energy region, and the similarities between both phases are consistent with DOS calculations, experimental measurements, and previous calculations from Ref. [48]. Above 5 eV, an increase in reflectivity is observed, indicating the presence of a band gap. Then, above 15 eV, the completion of transitions from the valence band to the conduction band occurs. The f-bands do not exert significant influence up to this energy level. Peaks at 20 and 23.5 eV arise from transitions between occupied f states and unoccupied states.

Lastly, as pointed out in previous sections, there is a significant difference observed between GGA-PBE and GGA-PBEsol functionals in calculating structural characteristics for various trifluorides. Therefore, optical features, namely reflectivity index, were also calculated using GGA-PBEsol functional with additional correction for d-electrons for DyF_3 and HoF_3. The results are presented in SFig. 12, while demonstrating an additional broad peak at 47 eV for DyF_3, whereas HoF_3 exhibits the same feature in calculations using both functionals.

4. Conclusions

The paper presented a comprehensive investigation of structural, mechanical, electronic, magnetic and optical properties of two possible phases of five rare earth trifluorides, namely GdF_3, TbF_3, DyF_3, HoF_3, and ErF_3. There are few important aspects obtained during the analysis and these might be summarized as following conclusions for each considered characteristic.

During the calculations with the common GGA-PBE functional, some problems were detected while calculating lattice constants for selected trifluorides. Namely, in the low pressure region, DyF_3, HoF_3,

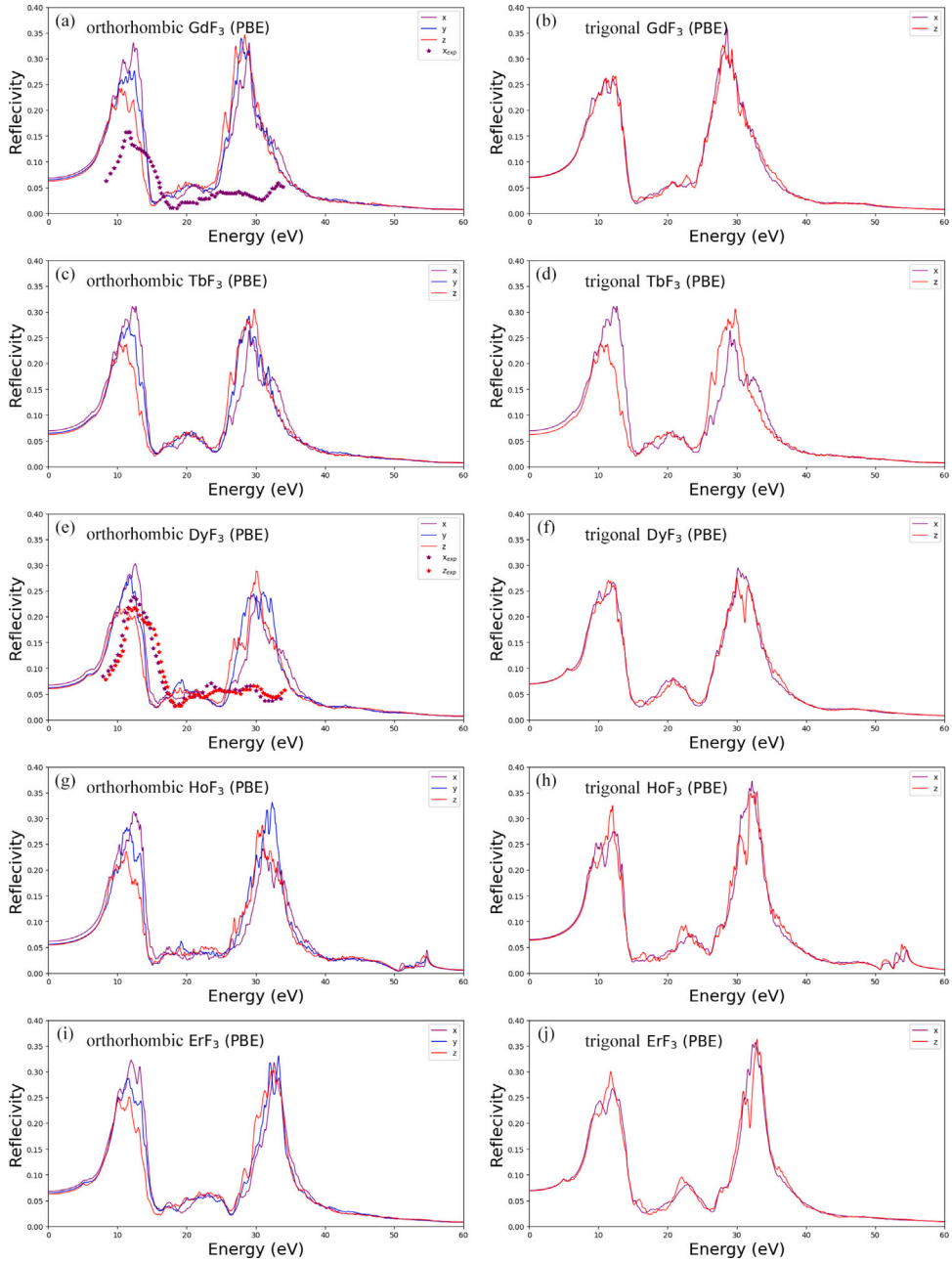


Fig. 6. Calculated using GGA-PBE functional reflectivity of orthorhombic and trigonal phases of GdF_3 (a, b), TbF_3 (c, d), DyF_3 (e, f), HoF_3 (g, h), and ErF_3 (i, j) crystals along with available experimental data from Ref. [56].

and ErF_3 exhibited nonphysical for these crystals behavior, typically associated with an anomalous negative linear compressibility effect. The resulting lattice parameters at 0 GPa differed significantly from experimental values. In order to overcome this issue, a number of U-corrections and functionals were tested to achieve the best fit to the experimental data. We found that the minimal deviation of unit cell parameters for GdF_3 and TbF_3 is provided by GGA-PBE functional and PAW potential, treating 4f-electrons of rare earth atoms as core states. In contrast, for DyF_3, HoF_3, and ErF_3, the approach with GGA-PBEsol functional and $+U_d$ correction is more suitable. We performed a search for the best U_d parameter for DyF_3 and HoF_3, resulting in a value of 2.3 eV for both of them. Moreover, we found that this parameter is suitable for ErF_3 and possibly for other subsequent elements in the lanthanide series.

Overall, the differences in the presented computational parameters required to treat various compounds from the rare earth trifluorides family advanced the understanding of these materials under pressure and highlighted the necessity of careful verification of parameters when calculating properties under such conditions. Besides, the study emphasizes the need for further, more detailed research to investigate the underlying causes of the observed behavior in the different considered compounds. Further, the main findings within the scope of the checked properties are briefly presented below.

Mechanical properties analysis revealed that both orthorhombic and trigonal phases are mechanically stable within 0–30 GPa range, as investigated through elastic properties. Besides, the elastic constants were thoroughly presented for all structures under consideration. Due

to the lack of experimental data, the calculated elastic coefficients and bulk moduli might be compared in further investigations.

While analyzing thermodynamic (energetic) characteristics we got that GGA-PBEsol functional provides proper description of all structures, in particular, the point of pressure induced PT. Besides, we confirmed that the trigonal structures with the $P\bar{3}c1$ space group is more energetically favorable than the hexagonal phase with the $P6_3/mcc$ over the 0–30 GPa pressure range. The obtained within selected computational scheme PT pressures between the orthorhombic and trigonal phases for considered structures are in good agreement with available experimental ranges.

Lastly, the electronic, magnetic and optical characteristics for the considered structures were presented in agreement with previously published DFT calculations and in partial agreement with quite old experimental optical spectra. The presented data can also serve as guidelines for future experimental investigations.

CRediT authorship contribution statement

Regina M. Burganova: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Zafari Umar:** Software, Methodology, Investigation. **Oleg V. Nedopekin:** Writing – review & editing, Validation, Project administration, Formal analysis. **Ilya V. Chepkasov:** Writing – review & editing, Validation, Methodology. **Irina I. Piyanzina:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.commatsci.2024.113428.

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